

IT SIMPLERA INSTITUTE

CYBER SECURITY ANALYST INTERNSHIP PROGRAM



WEEK 01 ASSIGNMENT

Cybersecurity Fundamentals & Security Concepts

Prepared by	Nabviya Qazi
Domain	Red Team Cybersecurity
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I. Introduction

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When we think about cybersecurity, it's easy to just picture hackers in hoodies typing in the dark. But after going through the material for this first week, it's clear that infosec is actually about building systems that can survive constant attacks. This report covers the basics: looking at how real companies got breached, understanding the CIA triad in practice, checking out some standard tools we'll be using, and figuring out how to set up a basic home network so it isn't completely exposed to the internet.

Task I :

Attack Name & Year	Organization	Attack Type	Description	Impact
Equifax Data Breach (2017)	Equifax	Application Vulnerability	Hackers exploited an unpatched Apache Struts web framework flaw to gain access to the corporate network	Stole sensitive personal data of 147 million people resulting in massive financial settlements
Colonial Pipeline Ransomware (2021)	Colonial Pipeline	Ransomware	Attackers used a compromised VPN password to deploy ransomware across the corporate network	Forced the shutdown of the largest fuel pipeline in the US causing severe regional gas shortages
SolarWinds Supply Chain (2020)	SolarWinds	Supply Chain Attack	Hackers compromised the software build process to inject a backdoor into official Orion software updates	Thousands of enterprise and government networks were compromised using trusted vendor updates
Target POS Malware (2013)	Target	POS Malware	Attackers used stolen third-party vendor credentials to push malware directly to point-of-sale devices	Stole 40 million credit and debit card numbers leading to a huge drop in consumer trust and profits
Sony Pictures Hack (2014)	Sony Pictures	Wiper Malware	Hackers destroyed internal servers using wiping malware and leaked unreleased movies and emails	Caused massive public relations damage and required a complete rebuild of the IT infrastructure

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Task 2:

Attack Name & Year	CIA Triad Component Violated	Explanation
Equifax Data Breach (2017)	Confidentiality	The attackers successfully bypassed security controls to read and extract highly sensitive private consumer records that were meant to be kept strictly secret
Colonial Pipeline Ransomware (2021)	Availability	The ransomware encrypted critical business systems meaning the company could not access their own data to safely operate the fuel pipeline
SolarWinds Supply Chain (2020)	Integrity	The official software update was maliciously altered before it reached the end users meaning the software was no longer in its trusted original state
Target POS Malware (2013)	Confidentiality	Customer payment card details were intercepted during live transactions meaning the financial data was exposed to unauthorized third parties
Sony Pictures Hack (2014)	Confidentiality and Availability	Private internal communications were exposed to the public violating secrecy and the IT systems were wiped making them totally unavailable for business use

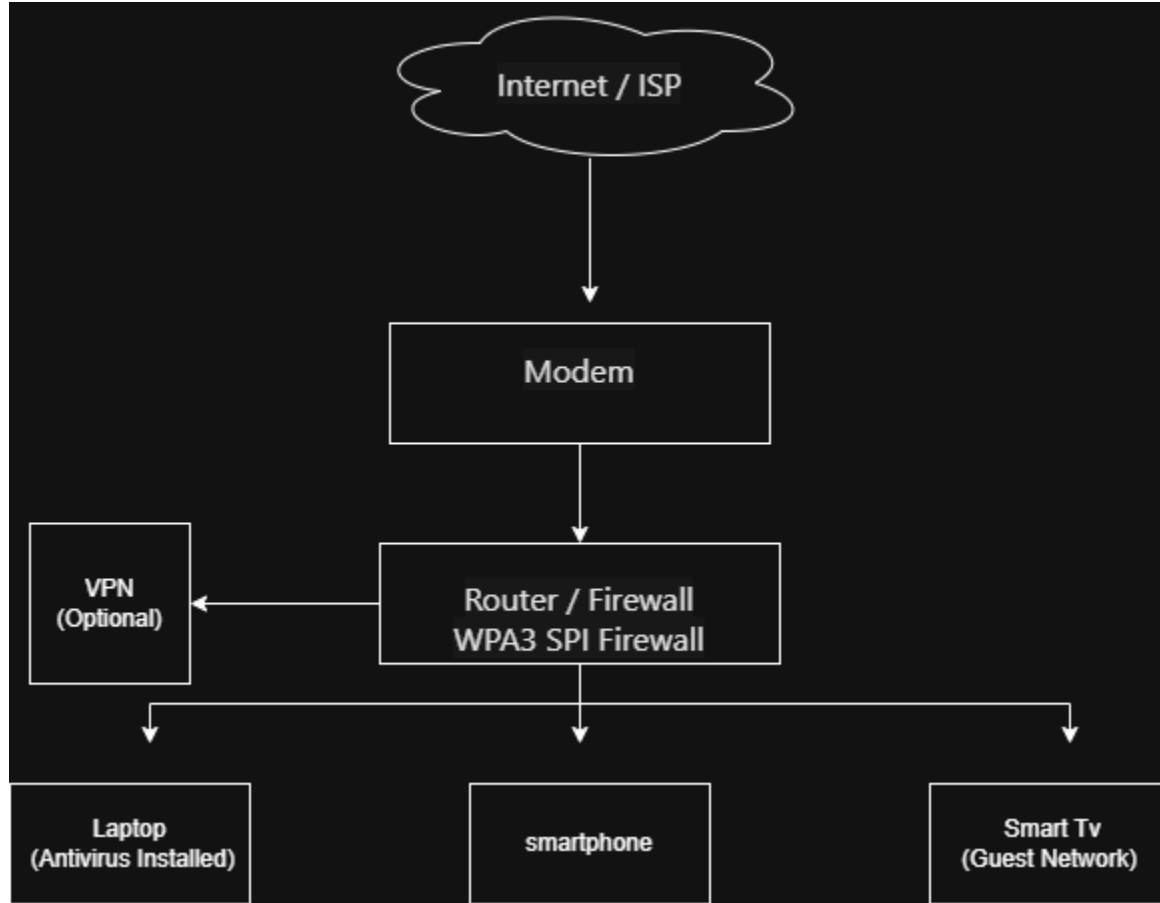
Task 3:

Tool Name	Purpose	Function	Real-World Use Case
Wireshark	Network Protocol Analyzer	It captures data packets traveling across a network in real time and displays them in a readable format for analysis	If a company suspects an employee laptop is secretly sending data to a hacker I would use Wireshark to capture and inspect that suspicious traffic
Nmap (Network Mapper)	Network Discovery & Scanning	It sends specially crafted packets to a target to discover active hosts open ports and running services on a network	Before conducting a penetration test I would use Nmap to scan a subnet to find any unauthorized devices or unpatched servers
Metasploit Framework	Penetration Testing / Exploitation	It contains a massive database of known vulnerabilities and exploits used to test and gain access to target systems	If Nmap shows a server is missing a specific patch I would use Metasploit to safely exploit that vulnerability on a test network to prove it needs fixing

Task 4: Create a secure home network diagram including Internet, Modem, Router/Firewall, devices, antivirus, firewall, optional VPN.

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Task 5: Write a one-page summary: What Does a Cybersecurity Analyst Do Every Day?

Before starting this internship, I thought a cybersecurity analyst spent all day looking at green code flying across a screen like in the movies. The reality is actually much more grounded, but honestly, just as interesting. A typical day is a mix of monitoring, investigating, and communicating.

The day usually starts with checking dashboards, specifically a SIEM (Security Information and Event Management) tool. The system generates hundreds or thousands of alerts overnight, and 95% of them are false positives—like someone just typing their password wrong three times. The analyst's job is to triage these, filtering out the noise to find the one alert that actually looks like a real threat.

If something looks suspicious, like a user account downloading 50GB of data to an unknown IP address at 2 AM, the analyst shifts into investigation mode. This means pulling logs from firewalls, Active Directory, and endpoints to build a timeline of what happened. Did the user's credentials get phished? Is it malware? You have to act like a digital detective to figure out if it was a real attack or just a false alarm.

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Besides reacting to alerts, analysts spend a lot of time on proactive work. This includes checking vulnerability scans to see if any new unpatched systems popped up on the network, writing documentation for how an incident was handled, and working with other IT teams to make sure firewall rules are actually doing what they are supposed to do. You aren't just defending; you are actively trying to break your own systems before the bad guys do.

Probably the most surprising thing I've learned is how much communication is involved. An analyst has to be able to explain a complex technical issue to a non-technical manager and convince them why a certain system needs to be taken offline or why a budget is needed for a new security tool. It's a balance of technical skills and office politics, which makes the job a lot more than just sitting alone at a computer.